Axolabs GmbH

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MDR_2439EXNA_RATTISSUE	EXECUTION OF THE PNA-HPLC-ASSAY FOR DETERMINATION OF 2439EXNA FROM RAT TISSUE (METHOD DEVELOPMENT REPORT)	
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	Name; Position	Date	Signature
PREPARATION:	**Thessa Jacob** **Scientist Bioanalytics**	06-Aug-2024	[redacted]
VERIFICATION:	**Markus Schuster** **Principal Research Associate**	06-Aug-2024	[redacted]

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1 RELATED DOCUMENTS

This instruction describes the execution of a PNA-HPLC-Assay based on the following documents:

- KU-RGL-SOP-002 (formerly: REG-102) – Methoden-Validierung im GLP/GCP-Bereich (*Method Validation in GLP/GCP Area*) with the following subordinate Instructions, Forms, Spreadsheet Applications:
  - KU-RGL-ESS-003 (formerly: REG-102-S03) – Method Validation Protocol PNA Assay - Solid Matrices
- KU-RGL-SOP-009 (formerly: REG-109) – Durchführung des PNA-HPLC-Assay (*Execution of the PNA-HPLC-Assay*) with the following subordinate Instructions, Forms, Spreadsheet Applications:
  - KU-RGL-ESS-012 (formerly: REG-109-S02) – PNA Assay Instruction and Protocol - Solid Matrices
  - KU-RGL-ESS-014 (formerly: REG-109-S04) – Preparation of HPLC Buffers & Stock Solutions
  - KU-RGL-INS-005 (formerly: REG-109-I01) – PNA-HPLC-Assay SST
- KU-RGL-SOP-010 (formerly: REG-110) – Herstellung von Standardlösungen mit einer Präzisionswaage (*Preparation of Standard Solutions Using a Precision Balance*) with the following subordinate Instructions, Forms, Spreadsheet Applications:
  - KU-RGL-ESS-017 (formerly: REG-110-S01) – Konzentrationsberechnung für eine Standardlösung (*Determination of the Concentration of a Standard Solution*)
- Development protocol in the project folder AN401.0-24 under the following link:

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The above mentioned documents describe the assay execution under nonGLP-relevant conditions. The parameters and results described in this document are generated from a nonGLP method development, which was conducted following the above mentioned SOPs.

2 DEFINITIONS AND ABBREVIATIONS

°C	Grad Celsius
ACN	Acetonitrile
AEX-HPLC	Anion-Exchange High-Performance Liquid Chromatography
CoA	Certificate of Analysis
CV	Coefficient of Variation
DNA	Deoxyribonucleic Acid
ed	Edition
EDTA	Ethylenediaminetetraacetic Acid
FDA	Food and Drug Administration
GLP	Good Laboratory Practice
HPLC	High Performance Liquid Chromatography
ICH	International Conference on Harmonization
IFU	Instructions For Use
LLOQ	Lower Limit of Quantitation
min	Minute
n.a.; N/A; N.A.	Not applicable
PBS	Phosphate Buffered Saline
PI	Principal Investigator
PNA	Peptide Nucleic Acid
PTO	Phosphorothioate
QC	Quality Control
rcf	Relative centrifugal force
RNA	Ribonucleic Acid
RT	Room temperature
SDS	Sodium Dodecyl Sulfate
siRNA	Small Interfering Ribonucleid Acid
SOP	Standard Operating Procedure
SD	Standard Deviation
SST	System Suitability Test(ing)
TCEP	Tris(2-carboxyethyl) phosphine hydrochloride solution
ULOQ	Upper Limit of Quantitation

3 ACTIVITIES TO BE PERFORMED

This method development report describes an anion exchange (AEX)-HPLC method with fluorescence detection that allows the specific detection of the analyte 2439exNA from rat tissue. 2439exNA is a dimer composed of two identical siRNAs, which are linked at their sense strands with a tetraethylene glycol linker, with the sequence depicted in Figure 1:

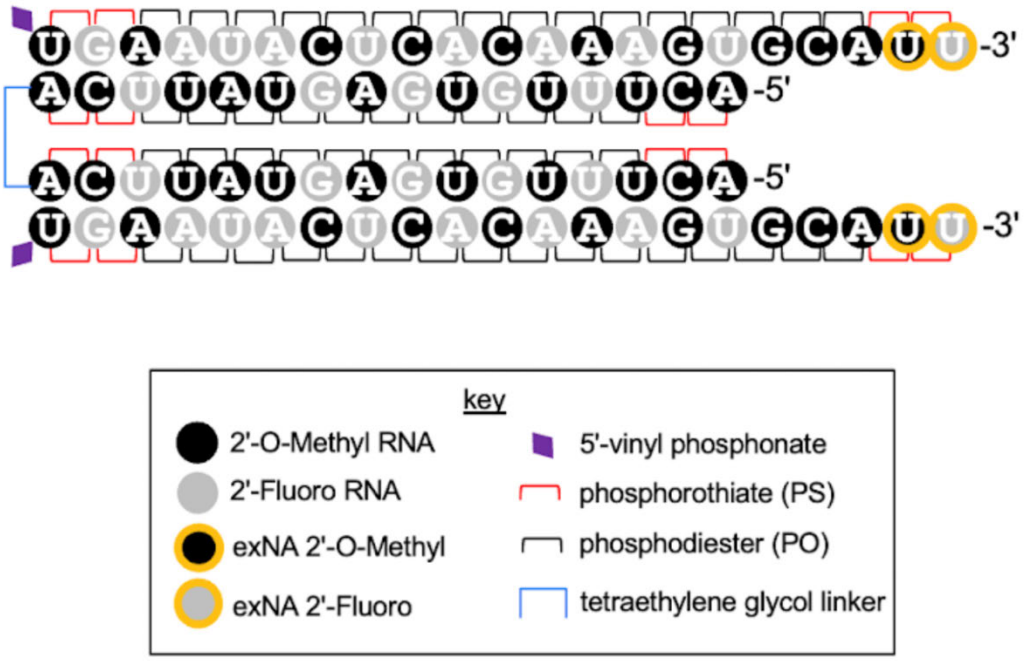

Figure 1 Structure and sequence of the analyte 2439-exNA

The assay is based on the specific hybridization of the antisense strands of 2439exNA to the complementary PNA-probe. The PNA-probe is N-terminally conjugated with an Atto425-fluorescent dye. The utilized PNA-BroadInst-01 is complementary to the antisense strand of 2439exNA and has the following sequence:

N-term Atto425-OO-cdc ttt gtg dgt d C-term

This method was developed without having any metabolites of 2439exNA

Figure 2 shows the HPLC-profile of 2439exNA in rat tissue.

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4 RESULTS FROM QUALIFICATION RUN IN RAT TISSUE

In this section, the raw data for the determination of calibration curve accuracy (Table 2), confirmation of LLOQ (Table 3), selectivity (Table 4), carry-over (Table 5), accuracy and precision (Table 6), freeze and thaw and short term stability (Table 9) and dilution integrity (Table 10) from 2439exNA in rat tissue are shown. The results were generated in one qualifying run. Calibration standard and QC samples were prepared by spiking 2439exNA into rat tissue. Samples were prepared according to the assay procedure described in Section 5.

During the qualification run, the calibration range was between 10.0 ng/g and 10,000.0 ng/g. All calibration standards were within ±15 % (±20 % at the LLOQ) of the nominal concentration.

Selectivity and Sensitivity

The LLOQ was confirmed at 10.0 ng/g in rat liver tissue. An analyte signal was detectable in blank tissue samples, but it did not violate the acceptance criteria, since the 2439exNA peaks at the LLOQ of 10.0 ng/g were at least 5 times the background signal in the analysis run. At the LLOQ, the mean back-calculated concentration was 104.0 % of nominal with a CV of 5.1 %.

The selectivity of the method was demonstrated by comparison of unspiked samples of the different rat tissues liver, kidney and brain with the same tissue samples spiked with 2439exNA at a concentration of the LLOQ. The signal for 2439exNA in the spiked samples gave an at least 5-fold higher response (peak area) compared to the blank samples.

Carry Over

Carry over tissue samples were run after the 10,000.0 ng/g calibration standards. There was no distinct analyte signal detectable in the carry-over samples after the 10,000.0 ng/g standards.

Accuracy and Precision

QC samples were prepared at 4 concentration levels which are covering the calibration curve range:

- at the LLOQ (QC-LLOQ): 10.0 ng/g

- at 3 times above the LLOQ (QC-Low): 30.0 ng/g
- at around 50 % of the calibration curve range (QC-Mid): 4,500.0 ng/g
- at least at 75 % of the upper calibration curve range (QC-High): 9,000.0 ng/g

The QC samples for the determination of the accuracy and precision of the method met the acceptance criteria with following results:

Liver

- 100 % of the QC samples at all concentration levels were within ±15 % (±20 % at the LLOQ) of their nominal value.
- The mean accuracies at the 4 concentration levels were 106.0 % (QC-LLOQ), 96.7 % (QC-Low), 100.2 % (QC-Mid) and 100.7 % (QC-High).
- The mean precisions at the 4 concentration levels were 0.9 % (QC-LLOQ), 2.2 % (QC-Low), 0.3 % (QC-Mid) and 0.4 % (QC-High).

Kidney

- 100.0 % of the QC samples at all concentration levels were within ±15 % (±20 % at the LLOQ) of their nominal value.
- The mean accuracies at the 4 concentration levels were 95.3 % (QC-LLOQ), 96.7 % (QC-Low), 99.0 % (QC-Mid) and 100.0 % (QC-High).
- The mean precisions at the 4 concentration levels were 3.7 % (QC-LLOQ), 1.6 % (QC-Low), 1.2 % (QC-Mid) and 0.2 % (QC-High).

Brain

- 100.0 % of the QC samples at all concentration levels were within ±15 % (±20 % at the LLOQ) of their nominal value.
- The mean accuracies at the 4 concentration levels were 98.7 % (QC-LLOQ), 97.1 % (QC-Low), 98.6 % (QC-Mid) and 99.2 % (QC-High).
- The mean precisions at the 4 concentration levels were 5.0 % (QC-LLOQ), 2.2 % (QC-Low), 0.6 % (QC-Mid) and 0.4 % (QC-High).

Freeze-Thaw and Short-Term-Stability

The freeze and thaw (one cycle) and short term stability of 2439exNA in rat tissue was successfully tested for up to 4 hours. Aliquots of the 2439exNA at the 2 QC concentration levels QC-Low and QC-High were stored at room temperature for 4 hours and analyzed in triplicate.

After 4 hours at room temperature, the mean back-calculated concentrations were at 94.1 % (QC-Low) and 93.2 % (QC-High) of the nominal value.

Dilution Integrity

The precision and accuracy of rat tissue sample dilution was evaluated using a QC standard prepared at a concentration level 100-fold (1,000,000.0 ng/g) above the ULOQ of 10,000.0 ng/g. This QC sample was analyzed and quantified after 1:1,000 dilution in 3 replicates. Samples were diluted with buffer solution (10 % ACN / 0.01 % Tween / 1x PBS / 1 mM TCEP).

- The mean accuracy was 95.5 % for the 1,000,000.0 ng/g dilution standard (1:1,000)
- The CV was 2.3 % for the 1,000,000.0 ng/g dilution standard (1:1,000)

Table 2 Calibration Curve Accuracy in rat liver tissue

2439exNA		Cal-Level-2		Cal-Level-3		Cal-Level-4		Cal-Level-5		Cal-Level-6		Cal-Level-7	
		10.0 ng/g		50.0 ng/g		500.0 ng/g		1,000.0 ng/g		5,000.0 ng/g		10,000.0 ng/g	
Run ID	Sample No.	Calculated 2439exNA Conc. ng/g	Acc. criterion +/-20 % of Nominal	Calculated 2439exNA Conc. ng/g	Acc. criterion +/-15 % of Nominal	Calculated 2439exNA Conc. ng/g	Acc. criterion +/-15 % of Nominal	Calculated 2439exNA Conc. ng/g	Acc. criterion +/-15 % of Nominal	Calculated 2439exNA Conc. ng/g	Acc. criterion +/-15 % of Nominal	Calculated 2439exNA Conc. ng/g	Acc. criterion +/-15 % of Nominal
AN401.0-24_01_AB01	5-10	10.7	107.0%	50.1	100.2%	492.2	98.4%	989.0	98.9%	5,032.7	100.7%	10,071.3	100.7%
AN401.0-24_01_AB01	76-81	9.7	97.0%	49.9	99.8%	492.6	98.5%	989.1	98.9%	5,021.3	100.4%	9,911.5	99.1%
Mean		10.20	102.0%	50.00	100.0%	492.40	98.5%	989.05	98.9%	5,027.00	100.5%	9,991.40	99.9%
SD (+/-)		0.71		0.14		0.28		0.07		8.06		113.00	
CV (%)		6.9%		0.3%		0.1%		0.0%		0.2%		1.1%	
N		2		2		2		2		2		2	
AN401.0-24_01_AB02	9-14	10.3	103.0%	48.7	97.4%	485.4	97.1%	978.7	97.9%	4,966.4	99.3%	10,024.6	100.2%
AN401.0-24_01_AB02	38-43	10.9	109.0%	49.3	98.6%	491.8	98.4%	985.4	98.5%	4,982.6	99.7%	10,086.0	100.9%
Mean		10.60	106.0%	49.00	98.0%	488.60	97.7%	982.05	98.2%	4,974.50	99.5%	10,055.30	100.6%
SD (+/-)		0.42		0.42		4.53		4.74		11.46		43.42	
CV (%)		4.0%		0.9%		0.9%		0.5%		0.2%		0.4%	
N		2		2		2		2		2		2	

Table 3 Confirmation of LLOQ in rat liver tissue

2439exNA		Cal-Level-1			Cal-Level-2		
		0.0 ng/g			10.0 ng/g		
Run ID	Sample No.	Peak Area [mV*min]	Calculated 2439exNA Conc. ng/g	Acc. Criterion ≤ 20 % of LLOQ	Peak Area [mV*min]	Calculated 2439exNA Conc. ng/g	Acc. Criterion +/-20 % of LLOQ
AN401.0-24_01_AB01	4, 5	0.0000	BQL	0.0%	0.0369	10.7	107.0%
AN401.0-24_01_AB01	75, 76	0.0000	BQL	0.0%	0.0327	9.7	97.0%
AN401.0-24_01_AB02	8, 9	0.0040	BQL	10.9%	0.0366	10.3	103.0%
AN401.0-24_01_AB02	37, 38	0.0023	BQL	5.9%	0.0389	10.9	109.0%
Mean		0.00158	n.a.	4.3%	0.03628	10.40	104.0%
SD (+/-)		0.00195	n.a.		0.00259	0.53	
CV (%)		123.6%	n.a.		7.1%	5.1%	
N		4	4		4	4	

Table 4 Selectivity in rat tissues liver, kidney and brain

2439exNA					Blank		QC-LLOQ	
					0.0 ng/g		10.0 ng/g	
Run ID	Date	Sex	Lot #	Sample No.	Peak Area [mV*min]	Acc. Criterion ≤ 20 % of LLOQ	Peak Area [mV*min]	Calculated 2439exNA Conc. ng/g
AN401.0-24_01_AB01 Liver	15-Jul-2024	M	RAT399893	13, 23	0.0000	0.0%	0.0361	10.5
	15-Jul-2024	M	RAT399894	14, 24	0.0000	0.0%	0.0367	10.7
	15-Jul-2024	M	RAT399895	15, 25	0.0000	0.0%	0.0365	10.6
AN401.0-24_01_AB01 Kidney	15-Jul-2024	F	RAT409712	16, 26	0.0000	0.0%	0.0320	9.5
	15-Jul-2024	F	RAT409711	17, 27	0.0000	0.0%	0.0334	9.9
	15-Jul-2024	F	RAT409696	18, 28	0.0000	0.0%	0.0308	9.2
AN401.0-24_01_AB01 Brain	15-Jul-2024	M	RAT409704	19, 29	0.0000	0.0%	0.0344	10.1
	15-Jul-2024	M	RAT409703	20, 30	0.0000	0.0%	0.0311	9.3
	15-Jul-2024	M	RAT409704/ RAT409703	21, 31	0.0000	0.0%	0.0346	10.2

M: Male, F: Female

Table 5 Carry-over in rat liver tissue

2439exNA		Cal-Level-2			Cal-Level-7			Carry Over		
		10.0 ng/g			10,000.0 ng/g			0.0 ng/g		
Run ID	Sample No.	Peak Area [mv*min]	Calculated 2439exNA Conc. ng/g	Acc. Criterion +/-20 % of Nominal	Peak Area [mv*min]	Calculated 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal	Peak Area [mv*min]	Calculated 2439exNA Conc. ng/g	Acc. Criterion ≤ 20 % of LLOQ
AN401.0-24_01_AB01	5, 10, 11	0.0369	10.7	107.0%	39.1767	10,071.3	100.7%	0.0000	BQL	0.0%
AN401.0-24_01_AB01	76, 81, 82	0.0327	9.7	97.0%	38.5550	9,911.5	99.1%	0.0000	BQL	0.0%
AN401.0-24_01_AB02	9, 14, 15	0.0366	10.3	103.0%	38.9667	10,024.6	100.2%	0.0000	BQL	0.0%
AN401.0-24_01_AB02	38, 43, 44	0.0389	10.9	109.0%	39.2054	10,086.0	100.9%	0.0000	BQL	0.0%

Table 6 Accuracy and Precision in rat liver tissue

2439exNA		QC-LLOQ		QC-Low		QC-Mid		QC-High	
		10.0 ng/g		30.0 ng/g		4,500.0 ng/g		9,000.0 ng/g	
Run ID	Sample No.	Calculated 2439exNA Conc. ng/g	Acc. Criterion +/-20 % of Nominal	Calculated 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal	Calculated 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal	Calculated 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal
AN401.0-24_01_AB01	23, 33, 43, 53	10.5	105.0%	29.2	97.3%	4,502.7	100.1%	9,034.1	100.4%
	24, 34, 44, 54	10.7	107.0%	28.3	94.3%	4,524.8	100.6%	9,097.7	101.1%
	25, 35, 45, 55	10.6	106.0%	29.5	98.3%	4,502.7	100.1%	9,060.0	100.7%
Mean		10.60	106.0%	29.00	96.7%	4,510.07	100.2%	9,063.93	100.7%
SD (+/-)		0.10		0.62		12.76		31.98	
CV (%)		0.9%		2.2%		0.3%		0.4%	
N		3		3		3		3	
AN401.0-24_01_AB02	17, 18, 19			28.5	95.0%	4,456.7	99.0%	8,951.5	99.5%
	33, 34, 35			29.4	98.0%	4,489.7	99.8%	8,951.6	99.5%
Mean				28.95	96.5%	4,473.20	99.4%	8,951.55	99.5%
SD (+/-)				0.64		23.33		0.07	
CV (%)				2.2%		0.5%		0.0%	
N				2		2		2	

Table 7 Accuracy and Precision in rat kidney tissue

2439exNA		QC-LLOQ		QC-Low		QC-Mid		QC-High	
		10.0 ng/g		30.0 ng/g		4,500.0 ng/g		9,000.0 ng/g	
Run ID	Sample No.	Calculated 2439exNA Conc. ng/g	Acc. Criterion +/-20 % of Nominal	Calculated 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal	Calculated 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal	Calculated 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal
AN401.0-24_01_AB01	26, 36, 46, 56	9.5	95.0%	29.1	97.0%	4,490.4	99.8%	8,970.8	99.7%
	27, 37, 47, 57	9.9	99.0%	28.5	95.0%	4,482.0	99.6%	9,002.8	100.0%
	28, 38, 48, 58	9.2	92.0%	29.4	98.0%	4,391.2	97.6%	9,013.6	100.2%
Mean		9.53	95.3%	29.00	96.7%	4,454.53	99.0%	8,995.73	100.0%
SD (+/-)		0.35		0.46		55.01		22.26	
CV (%)		3.7%		1.6%		1.2%		0.2%	
N		3		3		3		3	

Table 8 Accuracy and Precision in rat brain tissue

2439exNA		QC-LLOQ		QC-Low		QC-Mid		QC-High	
		10.0 ng/g		30.0 ng/g		4,500.0 ng/g		9,000.0 ng/g	
Run ID	Sample No.	Calculated 2439exNA Conc. ng/g	Acc. Criterion +/-20 % of Nominal	Calculated 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal	Calculated 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal	Calculated 2439exNA Conc. ng/g	Acc. Criterion +/-15 % of Nominal
AN401.0-24_01_AB01	29, 39, 49, 59	10.1	101.0%	29.6	98.7%	4,421.5	98.3%	8,925.9	99.2%
	30, 40, 50, 60	9.3	93.0%	28.4	94.7%	4,424.0	98.3%	8,966.4	99.6%
	31, 41, 51, 61	10.2	102.0%	29.4	98.0%	4,470.6	99.3%	8,899.7	98.9%
Mean		9.87	98.7%	29.13	97.1%	4,438.70	98.6%	8,930.67	99.2%
SD (+/-)		0.49		0.64		27.65		33.60	
CV (%)		5.0%		2.2%		0.6%		0.4%	
N		3		3		3		3	

Table 9 Freeze and thaw and short-term stability in rat liver tissue

2439exNA				QC-Low 30.0 ng/g		QC-High 9,000.0 ng/g	
Run ID	**Date**	**Sample No.**	**Time Point on wet ice**	**Calculated 2439exNA Conc. ng/g**	**Acc. Criterion +/-15 % of Nominal**	**Calculated 2439exNA Conc. ng/g**	**Acc. Criterion +/-15 % of Nominal**
AN401.0-24_01_AB01	15-Jul-2024	67, 71	1 Cycle -80°C 4h at room temp.	28.2	94.0%	8,383.6	93.2%
	15-Jul-2024	68, 72		28.8	96.0%	8,402.5	93.4%
	15-Jul-2024	69, 73		27.7	92.3%	8,365.3	92.9%
Mean				28.23	94.1%	8,383.80	93.2%
SD (+/-)				0.55		18.60	
CV (%)				2.0%		0.2%	
N				3		3	

Table 10 Dilution Integrity in rat liver tissue

2439exNA		1,000,000.0 ng/mL	
Run ID	Sample No.	Calculated 2439exNA Conc. ng/mL	Acc. Criterion +/-15 % of Nominal
Dilution reagent: 10% ACN/0.01% Tween/1xPBS/1mM TCEP, Dilution 1:1,000			
AN400.0-24_01_AB02	4	953,584.7	95.4%
	24	933,074.2	93.3%
	25	977,308.2	97.7%
Mean		954,655.70	95.5%
SD (+/-)		22,136.44	
CV (%)		2.3%	
N		3	
Dilution reagent: 4M Urea Dilution, 1:1,000			
AN400.0-24_01_AB01 1:1,000	93	916,352.6	91.6%
	94	891,019.8	89.1%
	95	920,018.3	92.0%
Mean		909,130.23	90.9%
SD (+/-)		15,790.83	
CV (%)		1.7%	
N		3	
Dilution reagent: 4M Urea, Dilution 1:100			
AN400.0-24_01_AB02 1:100	97	973,752.6	97.4%
	98	956,519.1	95.7%
	99	990,120.2	99.0%
Mean		973,463.97	97.3%
SD (+/-)		16,802.41	
CV (%)		1.7%	
N		3	
Dilution reagent: 4M Urea/1mM TCEP, Dilution 1:1,000			
AN400.0-24_01_AB02	5	637,778.4	63.8%
	27	548,316.8	54.8%
	28	494,576.2	49.5%
Mean		560,223.80	56.0%
SD (+/-)		72,339.82	
CV (%)		12.9%	
N		3	
Dilution reagent: LCMS-H_2O, Dilution 1:1,000			
AN400.0-24_01_AB02	6	865,737.5	86.6%
	30	839,825.4	84.0%
	31	808,896.6	80.9%
Mean		838,153.17	83.8%
SD (+/-)		28,457.32	
CV (%)		3.4%	
N		3	
Dilution reagent: 10% ACN/0.01% Tween/1x PBS, Dilution 1:1,000			
AN400.0-24_01_AB02	5	626,457.3	62.6%
	27	532,625.4	53.3%
	28	568,352.2	56.8%
Mean		575,811.63	57.6%
SD (+/-)		47,358.62	
CV (%)		8.2%	
N		3	

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6 RECORDS

The following records in addition to the above-mentioned Forms, Master or Spreadsheet Applications are required in this process:

- N/A

7 Change History

Edition, Section(s)	Change	Rationale
Ed 01	New document	Assay description after the method development